IEEE Access
Multidisciplinary : Rapid Review : Open Access Journal

Received 4 July 2025, accepted 13 August 2025, date of publication 20 August 2025, date of current version 8 September 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3600729

SURVEY

Customer Experience in E-Commerce: A Systematic Review of Metrics, Models, and the Role of AI

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ABSTRACT As digital commerce platforms grow increasingly intelligent and engaging, delivering a superior customer experience (CX) has become a strategic compulsion. Despite expanding interdisciplinary interest, clarity remains limited regarding how CX is conceptualized, measured, and transformed by emerging technologies. This systematic literature review synthesizes findings from 82 peer-reviewed studies published between 1980 and 2025 using the PRISMA 2020 framework to explore four key questions: how CX is defined in e-commerce, which dimensions influence it, what methods are used to evaluate it, and how technologies such as Artificial Intelligence (AI), agentic AI, and automation are used. The review reveals a shift from traditional, transactional, and usability-based interpretations towards dynamic, emotional, and co-created experiences. Trust, system-driven personalization, user autonomy, and customer engagement emerge as the four central influencing factors due to technological advances in AI and Agentic AI. Methodologically, CX research has evolved from survey-based models to structural equation modeling, sentiment analysis, and AI-integrated frameworks. Notably, intelligent systems are increasingly acting as co-creators of customer journeys, shaping real-time personalization and emotional resonance. This review offers a consolidated framework for understanding CX in the modern e-commerce context, provides actionable insights for platform designers, and proposes a future research agenda centered on factors and methods describing AI, agency, and emotional intelligence.

INDEX TERMS Customer experience, e-commerce, trust, user autonomy, personalization, customer satisfaction.

I. INTRODUCTION

Customer experience (CX) has become a primary focus in e-commerce and represents a critical driver of customer satisfaction, loyalty, and long-term business goals. Over the past four decades, the concept of CX has evolved from a narrow focus on post-purchase satisfaction to a dynamic, multidimensional construct shaped by digital touchpoints, emotional engagement, personalization, and adaptive technologies. In the context of e-commerce, in which interactions are often automated, fast-paced, and emotionally neutral by design, the intentional shaping of customer experience has emerged as both a strategic necessity and a competitive differentiator. Historically, the roots of CX lie in customer satisfaction theory, particularly the disconfirmation theory, which suggests that satisfaction arises from the gap between customer expectations and actual service performance [1]. During the 1980s and the 1990s, structured surveys using Likert scales were widely used to assess satisfaction as a post outcome metric. This measurement paradigm was extended by Parasuraman, Zeithaml, and Berry through the SERVQUAL model, which introduced five key service quality dimensions: tangibles, reliability, responsiveness, assurance, and empathy as evaluative components of perceived service quality. However, SERVQUAL was developed for traditional service settings and did not fully capture the richness or complexity of digital experiences [2].

The early 2000s marked a theoretical shift, as CX was redefined as a broader, holistic construct encompassing not only service delivery, but also emotional, cognitive, sensory,

The associate editor coordinating the review of this manuscript and approving it for publication was Venkateshkumar M.

and behavioral components across the entire customer journey. Verhoef et al. [3] emphasized the importance of both direct and indirect interactions in shaping the CX, while Gentile et al. [4] argued that value co-creation and affective experiences are essential dimensions of modern customer engagement. This re-conceptualization laid the foundation for customer journey mapping, touchpoint optimization, and cross-channel experience design.

In the digital commerce landscape, four factors have emerged as dominant influencers of CX: trust, personalization, user autonomy, and engagement. Trust is especially vital in online environments, where perceived risk and lack of face-to-face interaction may deter purchasing behavior [5], [6]. Personalization, enabled by algorithmic recommender systems and user analytics, allows platforms to tailor experiences based on preferences and past behavior, thereby increasing relevance and satisfaction [7]. Autonomy refers to the user's sense of control over their experience ranging from self-service navigation to configuration of delivery options, which significantly affects perceived ease of use and satisfaction [8]. Engagement, as described by Hollebeek et al. [9], captures the depth of cognitive and emotional involvement customers have with brands, often measured through behavioral signals such as clickstream data and interaction frequency.

Several frameworks and methodologies have emerged for managing and assessing these evolving dimensions. The experience quality (EXQ) model by Klaus and Maklan [10] provided a structured, multidimensional approach for evaluating CX in both digital and physical channels. Industry-standard KPIs such as Net Promoter Score (NPS), Customer Effort Score (CES), and Customer Satisfaction Score (CSAT) have also gained widespread adoption. More recently, the application of AI has transformed CX evaluation from retrospective survey-based models to predictive, real-time systems. Tools such as Aspect-Based Sentiment Analysis (ABSA), emotion recognition software, and AI-powered conversational agents allow businesses to detect and respond to customer needs dynamically and contextually [11].

The landscape of global commerce has undergone a radical transformation driven by the explosive growth of digital platforms, mobile-first behavior, and AI-driven personalization. In this evolving environment, customer experience (CX) has emerged as a critical source of competitive advantage in e-commerce, replacing traditional levers such as pricing or product differentiation with emotional engagement, perceived value, and technological fluency [11], [12]. CX is no longer confined to customer service or user interface design, and is now understood as a holistic, multi-touchpoint phenomenon that spans the entire customer journey, from product discovery and evaluation to post-purchase engagement and advocacy [13], [14]. This shift has prompted scholars to reconceptualize CX in e-commerce not merely as a transactional outcome, but as a subjective, co-created, and dynamic experience, influenced by a combination of platform design, user autonomy, trust mechanisms, and AI-enhanced personalization [15], [16].

A. WHY THIS REVIEW MATTERS

Although customer experience (CX) has garnered substantial attention in marketing, information systems, and digital commerce literature, the field remains fragmented and marked by conceptual ambiguity. There is an ongoing inconsistency in how CX is defined and understood across different time periods and contexts. Uncertainty also persists regarding the dimensions or factors that most critically influence CX outcomes in e-commerce environments. Moreover, the appropriate methodologies and models for accurately measuring and evaluating CX continue to be debated. The rapid emergence of technologies such as artificial intelligence, agentic systems, and automation has further transformed the nature of customer interactions: however, their specific impact on the structure and dynamics of CX remains underexplored. Considering the increasing complexity and intelligence of digital platforms, a comprehensive and systematic synthesis of this dispersed body of literature is both timely and essential.

B. RATIONALE OF THE REVIEW

CX is no longer a secondary concern limited to post-sale services, and now encompasses the entire customer journey, from discovery to decision-making, purchase, and loyalty. In this context, the CX has become a primary driver of differentiation, competitive advantage, and long-term customer retention. Despite the significant evolution of customer experience as a construct, the academic literature specific to e-commerce remains fragmented across disciplines, such as marketing, information systems, human-computer interaction, and AI. In recent years, scholars have begun to recognize the need to reframe CX as a dynamic, multi-dimensional, and co-created phenomenon involving both human and machine agencies [13], [14].

Many existing studies focus either on narrow performance metrics or on isolated stages of the customer journey, without synthesizing how CX is shaped across technologies, platforms, and customer behavior patterns. Furthermore, with the emergence of agentic AI and adaptive systems, new forms of human-technology interaction are reshaping expectations and experiential norms in e-commerce. This shift calls for a systematic synthesis of past and emerging literature to Clarify how CX has been conceptualized and evolved, identify core influencing factors, review the models and methods used in CX research, and understand how emerging technologies are transforming CX. Thus, this review is timely in its attempt to consolidate, map, and interpret the state of knowledge on CX within the e-commerce landscape from both conceptual and applied perspectives.

C. OBJECTIVES OF THE REVIEW

This systematic literature review aims to analyze, synthesize, and extend the existing body of research on CX in e-commerce using the PRISMA framework. Specifically, it addressed the following research questions:

- RQ1: How has customer experience been conceptualized in e-commerce? *(Theoretical coverage)*
- RQ2: What are the key dimensions and factors influencing customer experience on e-commerce platforms? *(Empirical relevance)*
- RQ3: What are the leading models, frameworks, and methodological approaches used to measure customer experience in e-Commerce? *(Methodological synthesis)*
- RQ4: How are emerging technologies, especially AI, agentic AI, and automation transforming customer experience in e-commerce ecosystems? *(Innovation and transformation angle)*

D. THEME OF THIS REVIEW

By synthesizing insights from 82 peer-reviewed studies published between 1980 and 2025, this review offers a comprehensive analysis of the evolution of the customer experience in e-commerce.

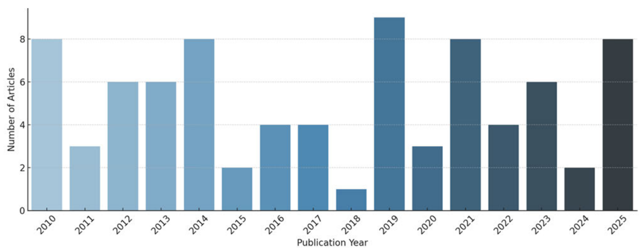

FIGURE 1. Distribution of selected research papers.

Figure 1 showing the distribution of the 82 selected studies by publication year. There's a clear rise in CX-focused research starting around 2014, peaking between 2019 and 2025 indicating the field's growing relevance in the AI-powered digital commerce era. It traces the progression of CX from its early focus on functionality and service quality to more recent interpretations that emphasize emotional engagement, agentic interaction, and intelligent system design.

Table 1 provides a grounded theme to conduct a systematic literature review (SLR). This study systematically maps the core measurable dimensions of the e-commerce CX. It also presents a timeline and classification of the key frameworks and analytical methodologies that have shaped CX research across different periods. Furthermore, the review examines the transformative role of emerging technologies, including artificial intelligence and automation, in redefining the architecture, perception, and delivery of the customer experience. Collectively, these contributions provide a robust conceptual foundation, practical reference for design and measurement, and forward-looking research agenda for scholars, practitioners, and digital platform developers seeking to optimize CX in increasingly intelligent and competitive online environments.

TABLE 1. Systematic review theme.

RQ	Theme	Literature study
RQ1	Conceptualization	Definitions, conceptual models, key dimensions (hedonic, utilitarian, cognitive, emotional).
RQ2	Factors/attributes influence CX	Factors like service quality, system quality, trust, satisfaction, perceived value, loyalty. Emphasis on online shopping behavior, user expectations, and e-commerce determinants.
RQ3	Methods and Measurement	Frameworks and models such as SERVQUAL, E-S-QUAL, EXQ, NPS, SEM, PLS, customer journey mapping, and evaluation techniques.
RQ4	Emerging technologies	AI, agentic AI, User autonomy, chatbots, AI personalization, Ethical AI,

E. KNOWLEDGE GAP ADDRESSED

While prior reviews have addressed customer satisfaction, usability, and personalization in e-commerce, no existing SLR comprehensively integrates the evolution of CX definitions, its multi-dimensional measurement, the role of AI/automation/agentic systems, and the emergent shift toward emotional and ethical CX. By addressing this gap, the current review aims to offer a robust conceptual foundation and actionable insights for both scholars and practitioners navigating the future of the digital customer experience.

II. METHODS

This systematic literature review was conducted to synthesize and evaluate how customer experience (CX) in e-commerce has been conceptualized, operationalized, and transformed, with a particular focus on emerging technologies. This review adheres to the **PRISMA 2020 framework** and employs a structured methodology across the selection, extraction, and synthesis phases.

A. ELIGIBILITY CRITERIA

Studies were included if they:

Focused on Customer Experience (CX) within e-commerce, m-commerce, and digital retail contexts.

Addressed at least one of the four research questions regarding:

- Conceptualization of CX
- Factors influencing CX
- Measurement frameworks and methods
- The impact of AI, Agentic AI, or automation on CX.

They were published between 1980 and 2025 in English and appeared in peer-reviewed journals, conference proceedings, book chapters, and reputable industry reports.

Studies were excluded if they:

- Focused solely on offline environments
- Discussed AI without linking it to CX.
- Lacked empirical, conceptual, or methodological contributions (e.g., blogs, editorials, promotional materials).
- Articles lacking customer experience relevance
- CX studies not involving digital/online context

B. INFORMATION SOURCES

The systematic search utilized a combination of academic databases, and supplementary sources:

- Academic Databases: Scopus, IEEE Xplore.
- Additional Scholarly Platforms: Google Scholar, ACM Digital Library, Emerald Insight, ResearchGate

C. SEARCH STRATEGY

Scopus Search Query:

((TITLE-ABS-KEY ("customer experience" OR "customer satisfaction" OR "customer feedback" OR cx OR "service quality")) AND TITLE-ABS-KEY (ecommerce OR commerce OR mcommerce OR "online shopping")) AND PUBYEAR > 1979 AND PUBYEAR < 2026 AND (LIMIT-TO (SRCTYPE, "j")) AND (LIMIT-TO (DOCTYPE, "ar")) AND (LIMIT-TO (SUBJAREA, "COMP") OR LIMIT-TO (SUBJAREA, "BUSI") OR LIMIT-TO (SUBJAREA, "ENGI") OR LIMIT-TO (SUBJAREA, "MULT") OR LIMIT-TO (SUBJAREA, "ECON") OR LIMIT-TO (SUBJAREA, "MATH") OR LIMIT-TO (SUBJAREA, "DECI") OR LIMIT-TO (SUBJAREA, "SOCI")) AND (LIMIT-TO (LANGUAGE, "English"))

IEEE Xplore Search Query:

("All Metadata": "customer experience" OR search All: "customer satisfaction")

Filters Applied:

Source type is Journals
Date range is 1980 – 2025

Manual paper additions:

- *("customer experience" OR "CX") AND ("e-commerce" OR "Online shopping") AND ("personalization" OR "AI" OR "automation" OR "trust" OR "engagement")*
- *("Customer Experience" OR "CX") AND ("e-commerce" OR "online shopping" OR "digital platforms" OR "digital retail") AND ("Artificial Intelligence" OR "AI" OR "Agentic AI" OR "Automation") AND ("factors" OR "measurement" OR "model" OR "framework" OR "impact")*

D. SELECTION PROCESS

Stage 1 – Identification
Records identified from Scopus (n = 3728)
Records identified from IEEE Xplore (n = 600)
Records identified from Google Scholar (n = 300)
Total records identified (n = 4628)
Stage 2 – Deduplication
Deduplication records removed (n = 1537)
Records after duplicates removed (n = 3,091)
Stage 3 – Title and Abstract Screening
Records screened (titles & abstracts) (n = 3,091)
Records excluded (n = 2,329)
Records moved to full-text screening (n = 210)
Stage 4 – Full-text Screening
Full-text articles assessed (n = 210)
Full-text articles excluded (n = 148)
Stage 5 – Final Inclusion
Final studies included in synthesis (n = 82)
Exclusion of records has carried out based on below:
Not meeting eligibility criteria
Insufficient methodological rigor
Not addressing any RQ

E. DATA COLLECTION PROCESS

A structured data extraction was developed capturing: Metadata (authors, title, year, source), Research objectives, CX conceptualization, Influencing factors, Measurement frameworks, AI/Agentic AI-related findings, Linkage to one or more of the four RQs. Data extraction and screening were performed using the Rayyan tool.

A coding sheet was developed to extract data including:

- Authors, year, title
- Research design (quantitative, qualitative, mixed)
- CX conceptualization (for RQ1)
- Dimensions or factors measured (for RQ2)
- Models/frameworks used (for RQ3)
- Role of AI, agentic systems, or automation (for RQ4)
- Country, sample size, method, and outcomes

Example variables coded:

- CX Dimensions: trust, usability, personalization, engagement
- Model Types: EXQ, TAM, UTAUT, customer journey mapping
- Technology Use: recommendation engines, chatbots, AI personalization

Figure 2 showing the frequency of CX dimensions identified across the reviewed articles. Key takeaways are Trust and Personalization are the most dominant dimensions. User Autonomy and Customer Engagement are highly represented, especially in recent literature. Also, Classical dimensions like Usability and Service Quality still appear but are less prominent.

F. DATA ITEMS

Extracted data included and Table 2, provides details on data items.

- Author, year, journal/source.
- Type of CX conceptualization.

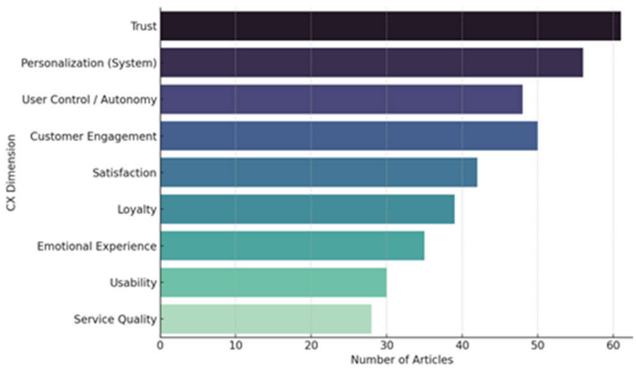

FIGURE 2. Frequency of CX dimensions in reviewed articles.

TABLE 2. Data item extraction.

Field	Description
Author, year	Source metadata
Research Focus	Empirical / Conceptual / Review
Dimensions of CX	trust, service quality, etc.
Methods Used	Surveys, models, text mining
Outcomes Measured	Satisfaction, loyalty, engagement, retention

- Identified influencing factors.
- Frameworks or models for CX measurement.
- Emerging AI/Agentic AI characteristics

G. STUDY RISK OF BIAS ASSESSMENT

Risk mitigated using multiple databases. Diverse methods across studies (survey, experimental, and behavioral) have been acknowledged. Selection bias was minimized via broad multi database coverage.

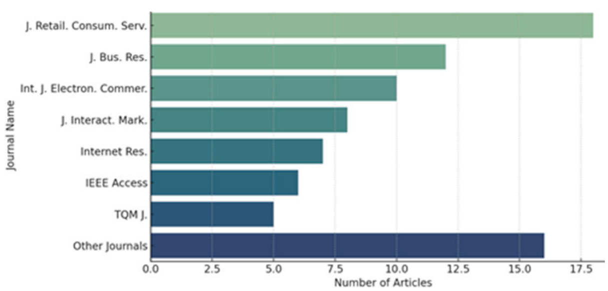

FIGURE 3. Selected studies by publication journal.

Publication bias reduced by including articles from multiple journals as dipicted in Figure 3.

H. EFFECT MEASURES

Since the review was primarily qualitative, patterns, themes, and factor frequencies were extracted. For AI-specific studies, emerging phenomena such as AI-induced stress and perceived autonomy, were highlighted.

I. SYNTHESIS METHODS

Thematic synthesis was employed to categorize the studies under the four RQs. The results are summarized in tables, thematic maps, and a conceptual framework. Frequency counts were used to highlight trends in CX factors and AI-related influences.

J. REPORTING BIAS ASSESSMENT

Three hundred research papers were manually selected from the literature and included along with other sources to mitigate reporting bias. The over-representation of AI's positive effects of AI in some industry reports was critically analyzed.

K. CERTAINTY ASSESSMENT

High certainty of traditional CX factors (trust, and personalization). Moderate certainty for AI-related factors due to limited research data. Emerging evidence for Agentic AI indicates the need for further research.

III. RESULTS

A. STUDY SELECTION

Figure 4 depicts step by step procedure followed for selection.

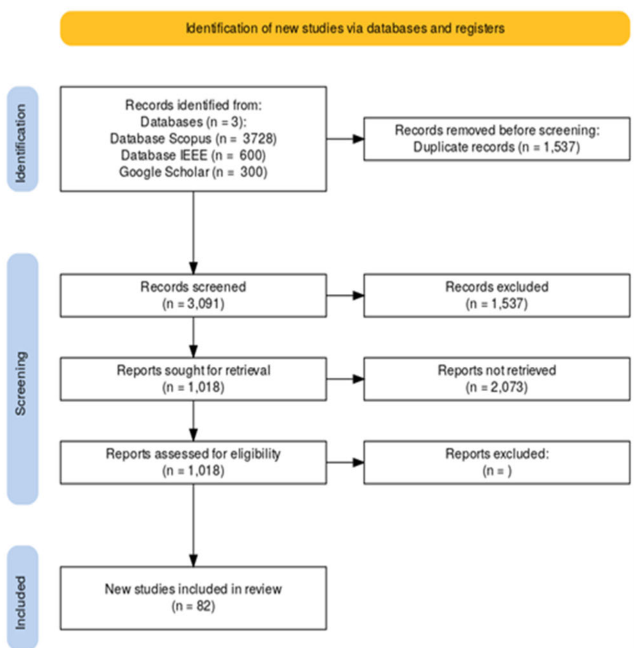

FIGURE 4. Study selection.

B. STUDY CHARACTERISTICS

- Studies spanned from 1980 to 2025.
- Covered topics from CX conceptualization to AI-enhanced customer interactions.
- Majority studies having mixed methods, machine learning, or qualitative modeling

C. RISK OF BIAS IN STUDIES

- No critical risk of bias detected.
- Most studies were peer-reviewed and applied sound methodologies

D. RESULTS OF INDIVIDUAL STUDIES

The final synthesis included 82 peer-reviewed journal articles that met eligibility criteria. These studies spanned 1980 to 2025, with a sharp increase in publication representation after 2014, particularly in areas related to digital personalization, AI applications, and customer-centric digital transformation. This section presents the findings of the four predefined research questions. These studies were evaluated for their relevance to the four research questions on CX in e-commerce.

RQ1: Conceptualization of Customer Experience in E-Commerce.

Scholarly literature contains many different ideas about what customer experience means in the context of e-commerce. Scholars have different definitions and focus on different aspects of this concept. When CX was first considered, it was often linked to the subjective and hedonic aspects of consumption, focusing on the pleasure, fantasy, and emotional responses people look for when they use goods and services. Table 3 provides definitions of customer experience by various researchers over time.

TABLE 3. CX definitions.

Author and Year	CX Definition
Holbrook and Hirschman (1982)[12]	An essentially subjective state of consciousness involving a variety of symbolic meanings, hedonic responses, and aesthetic criteria
Pine and Gilmore (1999)[13], [14]	Experience economy is the sale of memorable experiences to customers that surpasses commodities, products, and services
Verhoef et al. (2009)[3]	A multidimensional construct involves the customer's cognitive, affective, emotional, social, and physical responses
Lemon and Verhoef (2016)[15]	customer's Responses during entire purchase journey
Rose et al. (2011)[16]	Customer engages in cognitive and affective processing of incoming sensory information from the website, resulting in the formation of an impression in memory
Klaus and Maklan (2013)[10]	All direct and indirect encounters with organizations regarding Cognitive and affective behaviours
Kacprzak and Hensel (2023) [17]	Emotional, and behavioral responses of consumers with a company that take place through digital channels

Holbrook and Hirschman [12] proposed this view, which made it possible to see CX as a state that is fundamentally personal and subjective and is affected by symbolic meanings, hedonic responses, and aesthetic criteria. According to Pine and Gilmore [13], the idea of the progression of economic value made the point even stronger that experiences were more valuable than standard goods, services, and commodities. As the field has developed, an increasing number of people have agreed that CX in the digital world is multidimensional. These ideas consider that CX includes many different types of customer reactions, such as mental, emotional, behavioral, sensory, and social responses that occur when a customer interacts with an e-commerce business. For example, Verhoef et al. [3] clearly described the customer experience in retailing as a whole concept that includes the customer's mental, emotional, social, and physical reactions to the store. Similarly, Gentile et al. [4] pointed out that the purchase journey includes more than just functional rewards. It also has emotional, cognitive, relational, and sensory characteristics. Some definitions of CX also stress the temporal aspect, seeing it as the customer's journey with a company over time and at different points in the buying cycle.

The Internet and other digital technologies have changed the field of customer experience in many ways, adding new aspects and things to think about. Because e-commerce is conducted online, it is important to focus on making online transactions smooth and personal. CX is defined as how customers experience, feel, and act when they connect with an organization through digital channels, such as websites, social media, and mobile applications. In this digital setting, things such as how easy it is to use a website, how people talk to each other online, and the role of technology play a big part in shaping the whole experience.

A complete understanding of the customer experience that applies to modern e-commerce should include knowledge of how it is multidimensional and the online customer journey. It is also necessary to recognize how new technologies have had a significant effect on these events. Therefore, a full definition of e-commerce CX includes the mental, emotional, behavioral, sensory, and social responses of customers at all digital touchpoints during their contact with the brand as well as how these responses are affected by changes in technology.

Customer experience in e-commerce has been conceptualized in evolving ways over time, reflecting broader shifts in digital marketing, user psychology, and technology design. Early research emphasized functional and system-centric interpretations of CX, primarily grounded in the Technology Acceptance Model (TAM) [18] and Unified Theory of Acceptance and Use of Technology (UTAUT) [19]. These studies associated customer experience with perceived ease of use, system efficiency, and the intention to adopt digital interfaces.

As the field matured, a transition toward experiential definitions became evident, aligning with Service-Dominant Logic [20] and customer journey perspectives [21]. Here, CX was defined as a holistic construct involving not only usability but also emotional response, perceived value, and brand relationship across multiple touchpoints.

In more recent studies, particularly those post-2020, CX has been conceptualized as a dynamic and adaptive phenomenon, co-created by users and intelligent technologies [22], [23]. This study treats CX as an interactive and emergent construct influenced by real-time personalization, automated assistance, and user autonomy within digital ecosystems.

RQ2: Dimensions and Influencing Factors of Customer Experience

A multitude of factors, attributes, and parameters influences customer experience in the e-commerce domain. These factors span various aspects of the online shopping journey, from the initial interaction with the website to post-purchase engagement. Website design and usability play a pivotal role, with elements such as aesthetics, accessibility, and ease of navigation significantly impacting how customers perceive their online shopping experience [24]. A user-friendly interface and intuitive design contribute to a positive CX by enabling customers to easily find products and complete their purchases efficiently. Service quality and responsiveness are crucial determinants of CX in e-Commerce. Prompt and helpful customer support, efficient handling of inquiries and issues, and the overall quality of service provided by the online store directly influence customer satisfaction and perception of the experience [25]. Table 4 provides a list of the factors or attributes discussed in various studies. Trust and security are paramount in an online environment, where customers share personal and financial information [26].

TABLE 4. Factors influencing CX.

Factor	Description
Website Design and Usability[24]	Aesthetics, accessibility, ease of navigation, user-friendliness
Service Quality and Responsiveness[25], [27]	Prompt and helpful customer support, efficient handling of inquiries.
Trust and Security[26], [28], [29]	Secure payment gateways, clear privacy policies, trust signals.
Personalization [30], [31], [32]	Tailored recommendations, customized content
Product Information and Presentation[33], [34]	Quality of images, detailed descriptions, customer reviews
Post-Purchase Support[35], [36]	Timely delivery, clear communication, efficient returns
Social Interactions[37]	Social media integration, customer forums
Perceived Value and Pricing[38], [39]	Value received for the price paid
User Autonomy or User control[8], [40]	Increased user control enhances satisfaction and perceived usefulness, contributing to a better overall experience
Customer engagement[41]	Customer engagement in digital platforms influences brand relationships and loyalty

Factors such as secure payment gateways, clear privacy policies, and visible trust signals contribute to building customer confidence and a positive CX. Personalization and customization have emerged as key drivers of enhanced CX in e-commerce. Tailoring the online shopping experience to individual customer preferences, offering personalized product recommendations, and providing customized content can significantly enhance customer engagement and satisfaction. The way product information is presented, including the quality of images, detailed descriptions, and availability of customer reviews, also influences the CX by enabling informed purchase decisions [33].

During the entire customer purchase journey, product or service delivery and subsequent post-purchase engagement are important stages of the e-commerce journey that significantly impact overall CX [36]. A fulfilling post-purchase experience is provided to customers through prompt and reliable delivery, timely shipping and order communication, and the management of returns and exchanges. Aspects of social engagement and community, such as the incorporation of social media features, the availability of customer forums, and opportunities for interaction with other customers, can also influence CX by promoting a feeling of social validation and ownership. Finally, pricing and perceived value are also important factors. Customers base their assessment of their overall experience on how much they believe they are receiving the money they have spent [39].

These factors have an uneven effect on CX, and their relative significance varies based on specific scenarios, such as product type, market segment, and customer journey stage. Additionally, some factors might have a direct impact, while others attributes may apply their influence indirectly through mediating variables, such as trust or satisfaction. For instance, website content and security features can influence trust, which in turn affects the overall customer experience. By assigning priority to those that are most important to their audience and value proposition, e-commerce companies can effectively manage these influencing elements. These aspects must be continuously monitored and optimized to ensure long-term customer engagement and a positive CX [41].

The current study revealed four consistently emphasized factors that influence customer experience on e-commerce platforms:

1. Trust – Trust was closely associated with reliability, quality, security, transactional transparency, and the perceived integrity of automated systems and was featured in 74% of the studies [42], [43].
2. Personalization – In 68% of the reviewed articles, personalization was commonly framed as Website related ease of use, user-specific content delivery, recommendations and behavioral targeting [30], [32].
3. User Autonomy/User Control – In 59% of the studies, this factor refers to the extent to which users can navigate, filter, or configure their own digital experiences, especially in AI-mediated environments [8], [40]. This factor will emerge as the most influential factor in an AI-driven system environment. How much control of the configuration of payment methods, delivery options and the way to navigate can influence CX?
4. Customer Engagement – In 61% of the literature, engagement was measured through emotional agentic interfaces, and automation is transforming CX. How businesses engage customers during purchase and post purchase scenarios define their experiences. [9], [44],

TABLE 5. Frameworks, models, and methods for measuring CX.

Framework/Method	Description	Strength	Limitations	Relevant studies
Customer Journey Mapping	Visualizes the end-to-end customer interaction	Holistic view, identifies key touchpoints	Can be time-consuming, requires detailed data	[15], [21], [46], [47]
SERVQUAL/WEBQUAL/SITEQUAL	Measures service quality dimensions	Standardized, identifies areas for improvement	May not fully capture the online context	[2], [27], [48], [50]
System Usability Scale (SUS)	Measures perceived usability of websites	Quick, easy to administer	Focuses primarily on usability	[51], [52]
Customer Satisfaction Surveys	Collects quantitative data on customer perceptions	Direct customer feedback, quantifiable data	May lacks depth, potential for bias	[53]
Sentiment Analysis	Analyzes customer feedback from digital channels	Large-scale data, captures emotional tone	Relies on text analysis, may miss nuances	[54], [55], [56], [57]
Customer Experience Index (CEI)	Composite metric for overall CX	Provides a single, overall score	May not capture specific areas for CX	[58]

TABLE 6. Impact of emerging technologies on CX.

Technology	Application	Benefits	Concerns
Automation [59], [60]	Order processing, logistics, marketing campaigns, customer service (chatbots)	Streamlined operations, reduced errors, faster service	Potential lack of human touch, job displacement
AI [61], [62], [63]	Personalization (recommendations, content, pricing), Chatbots, Predictive Analytics	Enhanced personalization, improved efficiency, 24/7 support	Data privacy concerns, potential for bias
Agentic AI [64], [65]	Autonomous decision-making, proactive problem-solving, hyper-personalization	Real-time adaptation, reduced human oversight, enhanced efficiency	Ethical considerations, lack of transparency

These dimensions are frequently operationalized through a mix of subjective survey instruments and system-generated behavioral data, including clickstream analysis, sentiment from reviews, and dwell time tracking. Most recent studies have integrated multiple dimensions and highlighted their interdependencies, specifically between personalization, trust, and engagement.

RQ3: Models and Methods Used to Measure CX

In e-commerce, evaluation of customer experience is based on a wide range of frameworks, models, and approaches, each of which provides special insights and assessment techniques [45]. Table 5 describes the methods and models used over the years to measure CX. One useful framework for conceptualizing the entire customer journey with an e-commerce platform is the mapping of customer journeys, which helps identify important touchpoints and possible areas for development [15] This method provides a detailed view of the customer's experience across different stages, from initial awareness to post-purchase interactions [21], [46], [47]. Several popular models of service quality have been altered for use in e-commerce. SERVQUAL, a multi-dimensional service quality scale, has influenced modifications, such as WEBQUAL and SITEQUAL, which are intended to evaluate the perceived quality of e-commerce websites [48], [49].

These models assist in assessing various elements of digital services including information quality, reliability, and website design. The System Usability Scale (SUS) is widely used method that focuses on measuring the perceived usability of websites and online interfaces, which is a critical aspect of the e-commerce customer experience [50].

Customer satisfaction surveys remain a fundamental method for understanding customer perceptions and overall

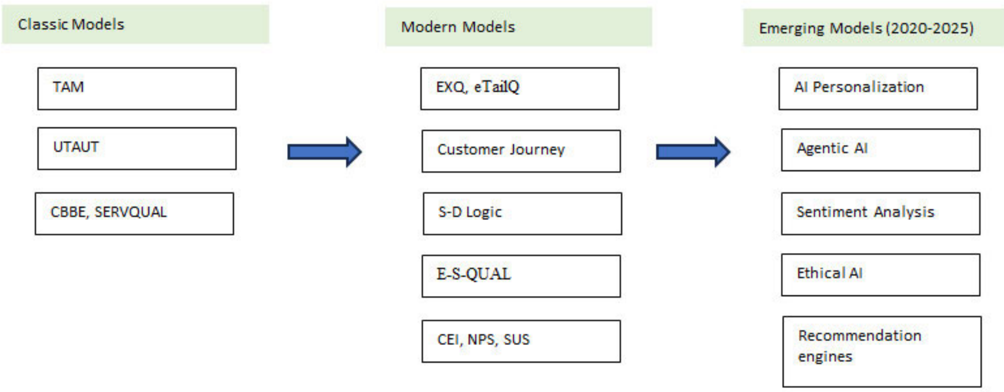

FIGURE 5. Methods and framework evolution in CX.

experience in e-commerce. These surveys can collect quantitative data on various aspects of the online customer journey, providing insights into customer satisfaction levels and areas for improvement. The Net Promoter Score (NPS) [52], which measures customer loyalty, and the Customer Effort Score (CES), which assesses the ease of customer interactions, are also commonly employed in e-commerce to measure CX.

Sentiment analysis is being increasingly used to understand customer experiences [53] owing to the rise of online reviews and social media. Online reviews, social media posts, and other digital channels that customers use to leave feedback can provide companies with useful information on how customers feel and what they think. The Customer Experience Index (CEI) is a composite metric that provides a full picture of the customer experience. This is done by examining many different parts of the customer journey [57].

Research analysis has revealed a significant shift in the frameworks and approaches used to study CX. Traditional models such as customer-based brand equity (CBBE), TAM, and UTAUT have been prominent in earlier research phases, particularly for understanding technology acceptance [18], [19]. Figure 5 shows the evolution of CX methods and frameworks. By the 2010s, researchers began adopting more comprehensive models, such as the Experience Quality (EXQ) Model, eTailQ, E-S-QUAL, and Customer Journey Mapping, which accounted for emotional, temporal, and interactional elements of CX [21], [65].

Studies have begun using AI-enhanced frameworks more frequently after 2020, integrating recommendation engines, emotion-aware automation, and real-time data analytics [66]. Sentiment analysis and deep-learning tools have emerged as popular methods for analyzing unstructured data from reviews and social media [11], [67]. Figure 6 shows the selected articles distributed according to the research method. Quantitatively, 59% of the studies used structural equation modeling (SEM) to validate the CX dimensions and their influence on outcomes such as satisfaction, loyalty, or conversion.

Mixed methods accounted for 22%, often blending interviews with survey or log data, while 19% relied on purely qualitative techniques such as case studies or thematic analysis.

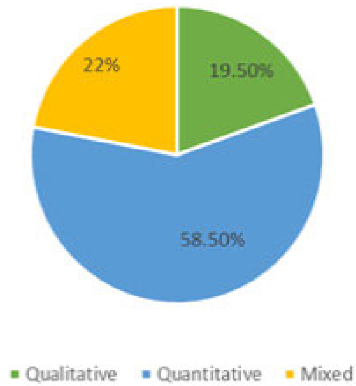

FIGURE 6. Research papers by applied methods.

RQ4: Role of Emerging Technologies in Shaping CX

The customer experience in e-commerce is changing dramatically owing to emerging technologies, especially automation, artificial intelligence (AI), and agentic AI [68], [69]. The key emerging technology impact topics are presented in Table 6. At the forefront of this change is AI personalization, which allows e-commerce companies to provide dynamic pricing strategies, highly customized product recommendations, and personalized content based on each customer's unique behavior and preferences [70]. By making customers feel appreciated, personalization increases customer engagement and cultivates enduring loyalty [71]. E-commerce customer service has been transformed by AI chatbots and virtual assistants, which offer round-the-clock personalized recommendations, order tracking, and instant answers to commonly asked questions [72]. Ultimately, these AI-powered solutions improve customer satisfaction by increasing productivity, reducing wait times, and offering scalability in customer service operations [73]. A more effective and seamless customer experience is facilitated by AI-driven automation of a variety of tasks such as order processing, logistics, and marketing campaigns [74].

Ultimately, this automation improves customer satisfaction by lowering manual labor, minimizing errors, and guaranteeing quicker service delivery. A more sophisticated type of AI is agentic AI, which is distinguished by its capacity to act and make decisions without direct human assistance [64]. Agentic artificial intelligence (AI) has the potential to transform customer experience (CX) in e-commerce by enabling hyper-personalization, proactive problem solving, and autonomous management of multiple customer journey elements, including dynamic pricing, inventory optimization, and real-time customer support [63].

Enhanced personalization, increased operational efficiency, round-the-clock customer service, cost savings, and proactive customer engagement are just a few advantages of integrating cutting-edge technologies [75]. However, there may be disadvantages in using AI and automation in e-commerce CX, which should be carefully considered. These include concerns about data security and privacy, the possibility of bias in AI algorithms, the impression that automated interactions lack human touch, the inability to comprehend intricate customer inquiries, and the constant requirement for system updates and maintenance [60], [61], [75].

E-commerce companies should prioritize data privacy, ensure transparency in AI usage, maintain a balance between automation and human interaction, continuously monitor and improve AI algorithms to prevent bias, and invest in training staff to collaborate with AI systems to harness emerging technologies for CX enhancement in an ethical and effective manner. To optimize the advantages of these technologies while minimizing any potential harm to customer experience, a human-centered approach is essential [60], [76], [77]. Although studies warn against the loss of human warmth or empathy, agentic systems, such as intelligent chatbots or virtual assistants, have been hailed for improving real-time responsiveness and customer autonomy. Workflows for customer service, content, and logistics are automated, which increases efficiency but also creates a sense of impersonality, especially in post-purchase service situations [78]. Only a small percentage of these studies have discussed the moral implications of technology-mediated CX. Instead of trustworthiness, explainability, or algorithmic fairness, most studies concentrated on functional outcomes (such as conversion rates and satisfaction), indicating a need for more critical and long-term studies.

IV. DISCUSSION

This systematic review analyzes findings from selected 82 studies spanning 1980–2025, and discusses how CX in e-commerce has evolved. As shown in Table 7, the discussion integrates insights from the four research questions to highlight theoretical evolution, practical implications, and emerging innovation challenges. The findings of this review provide a comprehensive view of how CX has evolved within the e-commerce literature over the past four decades. Rather than offering a static or singular definition, this research reveals that CX is an adaptive, multidimensional, and increasingly co-created phenomenon shaped not only by human interaction but also by technological advancements. This discussion interprets the results in relation to each research question, while also considering theoretical developments, methodological advancements, and future implications. A primary objective of this systematic review (**RQ4**) was to understand how emerging technologies, especially AI, are transforming customer experience. This chronological skew is not a methodological bias but a central finding of this review. It empirically demonstrates a definitive and rapid pivot in the scholarly community away from the foundational concepts that dominated earlier decades and towards the challenges and opportunities presented by AI, personalization, and automation.

TABLE 7. Key findings for research questions.

RQ	Key Findings	Supporting studies
RQ1	In addition to satisfaction, CX is multifaceted and comprises behavioral, emotional, cognitive, and sensory elements. Service quality has given way to engagement and personalization in CX. It is a comprehensive customer journey that spans several touch points rather than being transactional.	[10], [15], [16], [20], [80]
RQ2	Top influencing factors are trust, personalization, user autonomy, and customer engagement.	[9], [81], [82], [83], [84], [85]
RQ3	Make the transition from survey-based models to hybrid frameworks that integrate real-time data, behavioral analytics, and AI-generated personalization and recommendations.	[86], [87]
RQ4	AI improves CX through agentic decision-making, automation, and personalization. Risks, however, include reduced emotional engagement in the absence of human touch, stress caused through AI, and a perceived loss of control.	[88], [89], [90]

A. FROM USABILITY TO CO-CREATION: CX'S CHANGING CONCEPTUAL FOUNDATION

The conceptualization of CX has evolved from a transactional and usability-based principle into a comprehensive, multifaceted concept that includes behavioral, emotional, cognitive, and technological elements. As the cornerstone of digital CX, early research has mainly focused on interface quality, website usability, and service quality [25], [45].

This definition was broadened by the emergence of customer journey thinking to encompass social interactions, emotional states, and value co-creation. The conceptual development of CX reflects more general changes in information systems, marketing, and human–computer interaction. Experience was framed in early research in terms of system performance and usability [15], [18], and [19]. Such models do not consider the emotional, social, and longitudinal aspects of user experience, even though they are appropriate for early digital adoption contexts. A significant change was brought about by the introduction of Service-Dominant Logic [20]. It presented the idea of value-in-use, which subsequently influenced frameworks for customer journeys and the growing focus on experience as a process as opposed to a singular event [21]. However more recent research indicates that in today's algorithmic commerce environments, even co-created, emotion-centered definitions are inadequate. According to more recent conceptualizations, CX is a hybrid outcome of user intent and system intelligence that is agentically shaped [62], [90].

New theoretical frameworks that can explain how humans and machines collaborate to create perceived value, trust, and meaning in digital spaces are necessary in light of this shift. The integration of AI and agentic systems has impacted the conceptualization of CX in the past ten years, redefining it as an interactional phenomenon that is partially mediated by machines [63], [91], [92].

B. CUSTOMER EXPERIENCE DIMENSIONS: TRUST, USER CONTROL, PERSONALIZATION, AND CUSTOMER ENGAGEMENT

The focus on trust, personalization, autonomy, and engagement in the real world is increasing in customer experience (CX) research, but it also shows that these words are being used in new ways. Trust was measured using a simple group of questions on platform security, privacy and reliability. Variation in the application of trust reflects the evolving nature of Customer Experience (CX), especially with the integration of emerging technologies. In our systematic literature review, we specifically focused on the CX dimensions, measurement models, and emerging technologies within e-commerce.

As the field has evolved, trust has transformed from being primarily concerned with binary security perceptions (e.g., website security and privacy concerns) to more complex concepts, such as algorithmic transparency and AI-driven decision-making (ex. Personalization and recommendations). Our review aimed to highlight these shifts in CX factors by categorizing studies based on their focus on whether they are rooted in traditional security-based trust or more modern transparency-based trust influenced by AI and machine learning. An increasing number of studies are looking into how AI systems that are hard to understand may hurt trust, even when they work better (for example, by making recommendations faster). Similar to personalization, recommender systems have changed from static to context-aware and sentiment-sensitive algorithms [62], [93]. However, personalization without control of freedom may make people feel tired or like they are being manipulated [40], [94], [95].

The literature we examined shows that an increasing number of people are realizing that autonomy builds trust, which is why control features such as filter settings and opt-outs are so important for long-term CX. Engagement, which is often discussed about in terms of communication, support, emotions, and behavior, has grown to include cognitive, self-service and co-creative aspects. For example, when users provide feedback or use adaptive systems to change AI recommendations, they engage with the system [69]. This multifactorial view of CX makes us rethink how we measure it, and calls for mixed methods that combine behavioral data with subjective perceptions.

C. THE RISE OF INTELLIGENT MEASUREMENT AND METHODS

Over the years, customer experience measurements have transitioned from simple CSAT surveys to more complex systems that utilize real-time, intelligent, and analytical concepts. In the 1990s, frameworks such as SERVQUAL and CBBE gathered considerable attention and dominated academic and industrial discussions, focusing on service quality and brand equity. During the 2000s, eTailQ, E-S-QUAL, and the Net Promoter Score (NPS) were introduced and aligned with digital channels and loyalty. By the 2010s, KPIs such as Customer Effort Score (CES), Experience Quality (EXQ), and Customer Experience Index (CEI) were developed to work with customer journeys and Service-Dominant Logic, emphasizing co-creation and the full life cycle of customer experience. Organizations have begun integrating sentiment analysis, recommendation engines, and predictive analytics. In the 2020s, AI-powered personalization and complex AI frameworks were developed, enabling dynamic adaptation and trust-based measurement. Looking ahead, technologies such as LLMs, retrieval-augmented generation (RAG), and agentic AI are shaping empathetic real-time experience monitoring, creating a shift from reactive to strategic CX frameworks. As measurement tools have evolved, CX has also changed, moving from a reactive support function to a proactive and strategic contributor. This historical progression illustrates the growing complexity and importance of CX in shaping competitive advantages and long-term customer relationships.

D. TECHNOLOGY AS A MEDIATOR, NOT JUST A TOOL

The most important insight from the literature review is how technology change has impacted the customer experience in e-commerce. Technologies such as customer journeys, sentiment mining, AI, agentic interfaces, and automation are no longer treated as CX delivery tools but as active contributors to experience. This shift has profound implications for design, ethics, and strategy implementation. Studies have shown that when expert systems provide adaptive, personalized, and emotionally resonant responses, they enhance satisfaction and engagement. However, the same systems may reduce trust or perceived authenticity if users sense manipulation or a loss of control. Few studies have meaningfully addressed the ethical dimensions of these transformations, including data privacy, algorithmic bias, and fairness. As AI continues to invisibly shape experience, the field must develop experience models that account for human agency, transparency, algorithmic bias and customer value alignment.

TABLE 8. Future research recommendations.

Direction	Justification
Develop and validate cross-industry measurement frameworks	Overcome fragmentation and to test models across different industries
Study AI's psychological impact on customer trust and loyalty	To ensure ethical and sustainable AI use in CX
Explore agentic AI effects beyond chatbots	To understand advanced AI interactions
Conduct longitudinal studies	To assess AI-driven CX over time
Integrate emotional analytics into CX frameworks	To better capture affective dimensions of CX
Develop algorithmic transparency, bias, and fairness in selecting CX measurement models	To asses Trust, User autonomy and Personalization dimensions where algorithmic transparency and bias are included.
Sociocultural and Cross-Cultural CX Variations	To investigate how cultural dimensions may moderate the importance and perceptions

E. INTEGRATIVE PERSPECTIVE

Advancements in emerging technologies have transitioned from a support tool to a central agent in shaping the customer experience (CX) across digital commerce. Initial applications such as survey tools, rule-based recommendation engines, automated communications, and basic chatbots were primarily designed to optimize operational efficiency and reduce service costs. With the advancement of machine learning and natural language processing, AI has begun to enable scalable personalization based on behavioral and transactional data. Contemporary frameworks integrate AI and RAG, allowing systems to infer intent, adapt dynamically, and respond in contextually appropriate ways. This evolution has positioned AI not merely as an automation layer but also as a contributor to the customer journey, participating in continuous interactional feedback loops. To avoid over-automation and cognitive overload, recent guidelines have emphasized design principles that prioritize user empowerment and transparency. As such, AI-driven CX is increasingly being understood as a co-constructed process involving human cognition, algorithmic decision-making, and cross-channel interaction. To remain theoretically relevant and practically applicable, CX research must shift toward real-time, adaptive, and emotionally intelligent measurement frameworks. Furthermore, ethical considerations, cultural variance, and the integration of behavioral, cognitive, and affective data are essential for designing inclusive and responsible AI-mediated experiences. These developments mark a paradigm shift wherein customer experience is not passively delivered, but dynamically co-authored between users and intelligent systems.

V. CONCLUDING REMARKS

Over the past three decades, customer experience in e-commerce has grown from an emerging idea to a central principle guiding digital business strategies. This systematic review traced evolution through definitions, influencing factors, measurement frameworks, and infusion of emerging technologies. The key insights can be summarized as follows.

- **Definition Evolution:** The concept of CX has been defined in progressively richer ways, from a transactional, usability-focused construct to a holistic, emotional, and dynamic multi touchpoint journey that evokes positive cognitive, emotional, and social responses. The e-commerce context added nuances, such as the importance of online interface design and trust-building, but ultimately reinforced that CX is holistic and personal.
- **Influencing factors of CX:** Numerous factors influence e-commerce CX, and successful experiences result from aligning functional elements (usability, reliability, and product value) with emotive elements (aesthetics, brand image, and customer care) and social elements (community and trust). Organizations must manage all dimensions, and a deficiency in one (e.g., engagement or lack of trust) can undermine the rest. The literature consistently shows that trust, personalization, user autonomy or control and customer engagement are among the top drivers of positive CX in e-commerce.
- **Frameworks & Measurement:** Researchers have developed strong conceptual frameworks and practical tools to understand and measure CX. A methodological shift from traditional surveys to structural modeling, sentiment analysis, and AI-integrated behavioral analytics. A critical review of these tools suggests that no one-size-fits-all metric exists; instead, organizations should adopt a balanced scorecard of CX metrics (satisfaction, NPS, resolution time, emotional indicators, etc.) and utilize new techniques (text mining, real-time feedback) for a comprehensive measurement.
- **Technological Transformation:** Emerging technologies, such as AI, machine learning, chatbots, voice assistants, augmented reality, and Agentic AI, are reshaping the e-commerce landscape. They hold the promise of elevating CX to new heights of personalization, proactivity, and immersion. AI can "make e-commerce experiences more personalized, efficient, and engaging." for example, by anticipating the requirements or

providing human-like services at the scale. However, these technologies must be implemented carefully to maintain trust and empathy in customer interactions. The next phase of CX will likely be defined by how well organizations blend technology-driven capabilities with human-centered design.

- **Impact on Business Performance:** A recurring theme is that improving CX is not just a feel-good endeavor; it has concrete payoffs. Satisfied, emotionally engaged customers tend to buy more, stay loyal longer, and share positive experiences, fueling organic growth. Conversely, a poor experience (e.g., a frustrating website or bad AI chatbot interaction) can quickly lead to lost sales and reputational damage in an era where negative word-of-mouth (WOM) spreads quickly. Thus, mastering CX in e-commerce is imperative. The growing role of AI and automation is not just as tools, but also as active co-creators of the customer journey, with both opportunities and risks for trust, autonomy, and satisfaction.

Overall, CX in e-commerce has become an interdisciplinary, technology-mediated, and emotionally intelligent domain that requires ongoing adaptation from both scholars and practitioners. This review demonstrates that customer experience in e-commerce is no longer confined to static touchpoints, it is a fluid, co-created, and intelligent process shaped as much by technology as by emotion. As we move into an AI-driven era, the future of CX depends on our ability to create meaningful, transparent, and empowering interactions between humans and digital platforms.

In conclusion, the journey of e-commerce customer experience knowledge over 40 years shows a clear trajectory towards more integration, sophistication, and customer-centricity. Early e-commerce was about obtaining the basics of online shopping; today, it is about delighting customers in many ways across an omni-channel customer journey. As we stand in 2025, businesses have more tools and understanding than ever to engineer great experiences, from psychological insights into what customers value, to AI tools that can help deliver those values at scale. Successful organizations are those that continuously adapt to evolving customer expectations, leverage technology thoughtfully, and never lose sight of the human at the center of the experience. Future research should undoubtedly continue to unravel new dimensions of CX, but the core principle remains: customer experience is the core of e-commerce success, and understanding its facets and drivers is essential for success in the digital marketplace.

A. THEORETICAL CONTRIBUTIONS

- Updated Conceptual Foundations: This review charts the theoretical evolution of CX from SERVQUAL/CBBE frameworks to experiential, S-D logic, and AI-augmented perspectives, offering a consolidated view of the CX literature over 40 years.
- CX Dimensional Model for E-Commerce: A new synthesis of four precise, measurable dimensions; Trust, Personalization, User Autonomy, and Customer Engagement as core building blocks of CX.
- Framework Evolution Mapping (1980–2025): A visual and analytical mapping of CX models across three eras; Classic, modern, and emerging, showing paradigm shifts in constructs and methods.
- Technology Centric CX Theory: This review integrates the literature on agentic AI, sentiment mining, and predictive personalization, expanding the CX theory into AI-driven domains.

VI. LIMITATIONS

This review provides a comprehensive synthesis of customer experience (CX) research in e-commerce, but the following limitations should be acknowledged.

Inclusion Scope: The study reviewed 82 selected articles using a strict PRISMA process; however, the primary focus was largely on peer-reviewed journal articles in English, potentially excluding significant insights from industry reports, conference proceedings, or non-English sources. The data selection scope was for Scopus, IEEE, and Google Scholar, and the study did not consider the WOS database.

Temporal Distribution: Current study included research from 1980 to 2025, but the last ten years have been disproportionately represented, particularly in emerging technology related contexts (AI, automation). This reflects the current interest, but may create an importance bias toward innovation-focused literature.

Subject to interpretation: While coding and synthesis were systematic, the interpretation of thematic dimensions (e.g., autonomy, and engagement) involves qualitative judgment. Despite inter-reviewer checks, this introduces potential subjectivity into the classification.

Pure Technological Focus: This review predominantly focuses on emerging technologies but does not deeply explore CX variation across demographics, such as age, culture, or digital literacy, an area that should be considered for future discoveries. Exploring variations in CX across age, culture, and digital literacy would require a separate, in-depth research study that focuses on the sociocultural dimensions of CX. These factors are qualitative in nature and might necessitate cross-disciplinary approaches involving sociology, psychology, and behavioral science, which were not the primary focus of this review. Detailed analysis of geographic distribution and industry-specific CX models was beyond the scope of this systematic review, and therefore the findings represent broad trends rather than context-specific certainties.

Measurement Heterogeneity: There is no universally accepted gold standard for measuring customer experience (CX), leading to significant heterogeneity in both methodology and interpretation across the literature. Existing CX measurement approaches span a wide spectrum, from traditional psychometric instruments such as Likert-scale surveys (e.g., CSAT, NPS, and CES) to behavioral analytics derived from clickstream data, session recordings, and A/B testing.

More recently, researchers have incorporated AI-driven techniques, including sentiment analysis, emotion detection, and topic modeling of unstructured text (e.g., reviews, social media, and chatbot transcripts). These diverse methods reflect different underlying assumptions. While survey-based models typically capture perceived experience retrospectively, behavioral and AI-based approaches aim to infer experience dynamically and, in some cases, pre-emptively. Moreover, emerging studies often adopt hybrid analytics frameworks that, integrate structured and unstructured data sources to triangulate CX insights. This diversity, while methodologically rich, presents challenges for cross-study comparability, especially when definitions of constructs, such as satisfaction, trust, or engagement, are operationalized differently. Additionally, measurement techniques vary across industries, cultures, and digital maturity levels, making standardization difficult. In our review, we categorized the most commonly used measurement tools into three broad clusters: (1) attitudinal measures (e.g., Likert scale surveys), (2) behavioral/transactional indicators (e.g., clickstream, usage frequency), and (3) computational intelligence methods (e.g., AI models, LLMs, and emotion detection). Recognizing this heterogeneity is critical for evaluating the generalizability and robustness of CX findings, and underscores the need for future research to work toward interoperable metrics, standardized ontologies, and context-aware model adaptation.

VII. FUTURE RESEARCH

Based on the gaps identified in the literature and current technological trends, Table 8 shows the future research focus:

DECLARATION OF CONFLICTING INTERESTS

The authors declare no potential conflicts of interest regarding the research, authorship, or publication of this article.

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